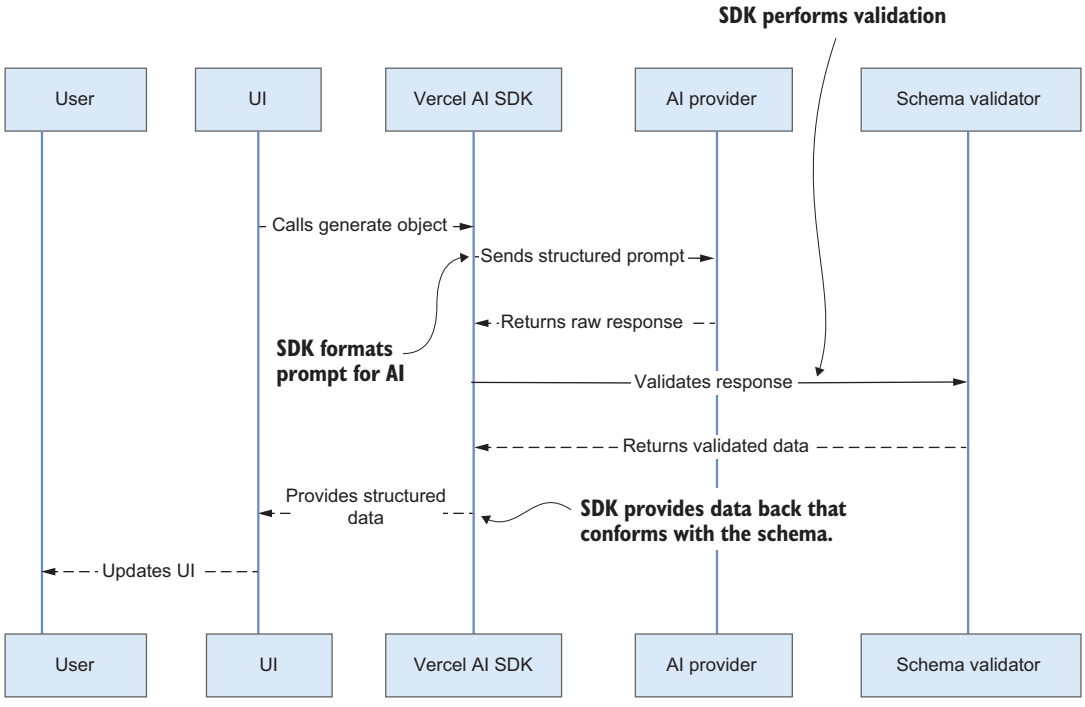


Figure 4.11 This sequence diagram illustrates the process of generating structured data from an AI model using the Vercel AI SDK. The flow begins with a user action and ends with the display of validated, structured data in the user interface. Key components include the React component, Vercel AI SDK, AI model, and schema validator.

This process ensures that the data received from the AI model is not only relevant but also follows a predefined structure, making it easier to integrate into applications. Let's look at the key techniques for generating structured data from AI responses.

4.3.2 Techniques for generating structured data from AI responses

Before we get technical, I explain some techniques we can use to instruct the AI model to respond in a certain format. To effectively generate and handle structured data from AI responses, we can employ several techniques, such as

- *Prompt engineering*—This involves crafting precise, detailed prompts that guide the AI to produce structured outputs like JSON objects with specific fields.
- *Output parsing*—Another way is by using regex patterns or JSON parsing to extract structured data from AI responses. This is trickier but still viable in certain scenarios by allowing flexibility in AI outputs while still obtaining structured data.
- *Function calling*—This is a feature provided by certain AI providers like OpenAI for requesting specific data structures. With this method, we define a function schema that the AI model must adhere to in its response.